Fail Closed

No approval, no signature.

An independent co-signer for crypto wallets. Unusual withdrawals run in an isolated sandbox before anything is signed.

AI on Vultr Serverless Inference · the demo runs on a laptop (the Vultr VM was blocked by account review)

THE PROBLEM

Two real attacks. Valid signatures, wrong withdrawals.

Binance · 29 Oct 2026

Hot wallet delegation drain (EIP-7702)

- About $6.3 billion left hot wallets on about 10 chains.
- One Ethereum transaction carried 50 EIP-7702 authorizations for 50 consecutive nonces; one matched and handed the wallet to the attacker's contract (Blockaid).
- Binance says it found no evidence of a private key leak; the investigation is ongoing.

Mercado Bitcoin · 2 Nov 2026

Multi-chain hot wallet drain

- About $38–49M (estimates vary), in plain signed transfers.
- Attacker addresses on at least 10 chains; hundreds of tokens (about 185 on Ethereum alone).
- Plain transfers, not a smart-contract exploit (Hypernative).

Fail Closed is designed to reject this class of attack. We do not claim it would have stopped these incidents.

THE COST NOBODY COUNTS

After the attack, everyone waits.

- Both exchanges paused withdrawals **for all customers**.
- Honest customers pay for the attack too: their money is stuck.
- A full stop is the only tool when nothing narrower exists.
- **A guard should stop the attacker, not the business.**

Fail Closed is designed to reject this class of attack. We do not claim it would have stopped these incidents.

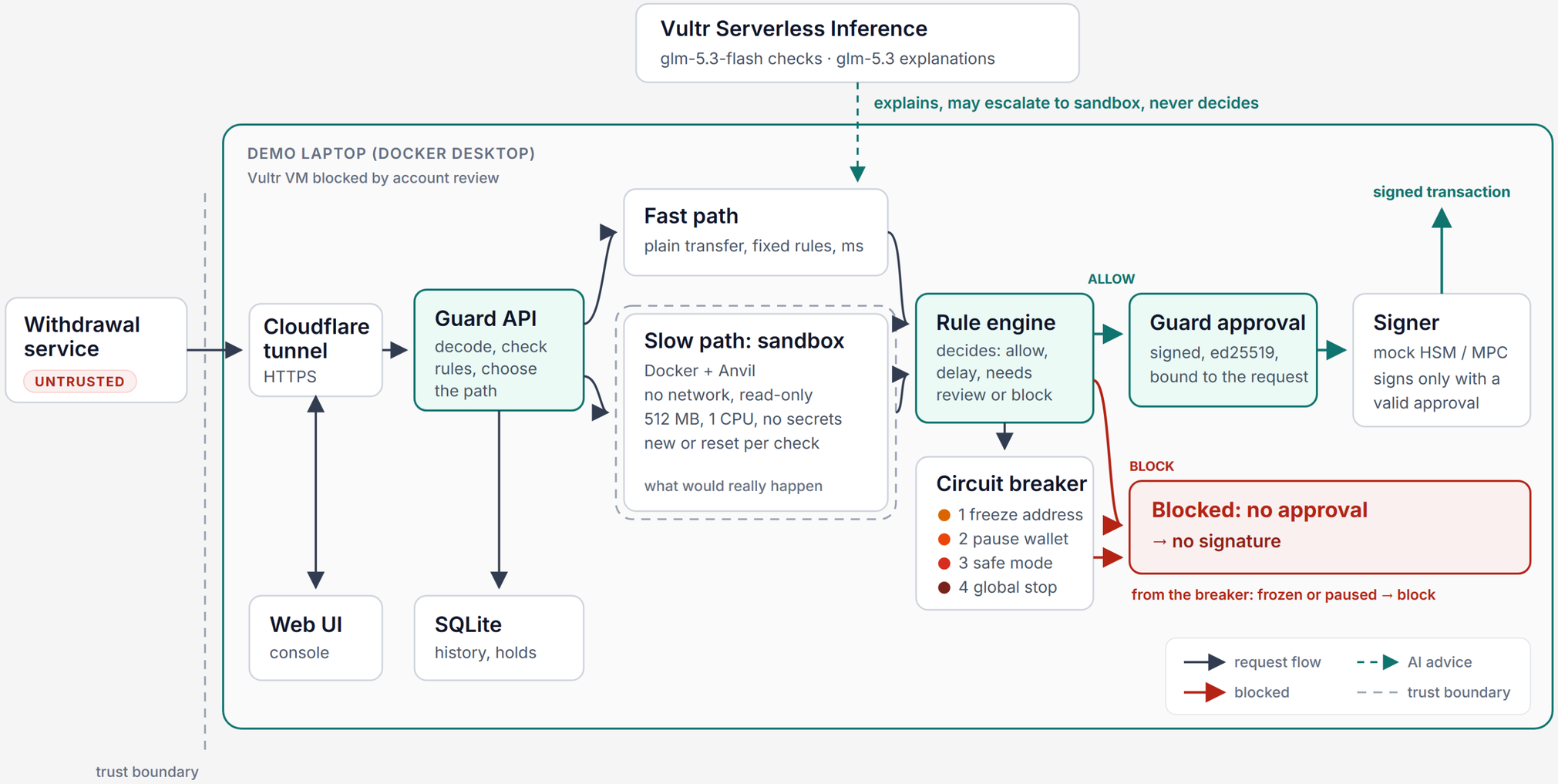
THE IDEA

Check what a signature will do, not who asks.

- **An independent co-signer.** The signer signs only with a Guard approval for that exact request.
- **Unusual requests run first** in an isolated sandbox: no network, no secrets, fake funds.
- **Fixed rules decide.** The AI explains in plain words. It may ask for a deeper check, never decides.
- **Fail closed.** Any error, timeout or empty result: no approval, no signature.

Fail Closed: how a signing request is checked

No approval, no signature. Fixed rules decide; the AI only explains.



WHERE FAIL CLOSED FITS

The enforcement point exists. Fail Closed is a prototype of the decision logic.

Screening before signing

Threat intelligence and pre-signing checks already exist as products: **Blockaid Cosigner**, **Hypernative Transaction Guard**.

Signer hooks

DFNS Validation Gate, **Fireblocks API Co-signer Callback Handler**, **Fordefi Custom API Approver** give the enforcement point. The customer writes the decision logic.

Fail Closed

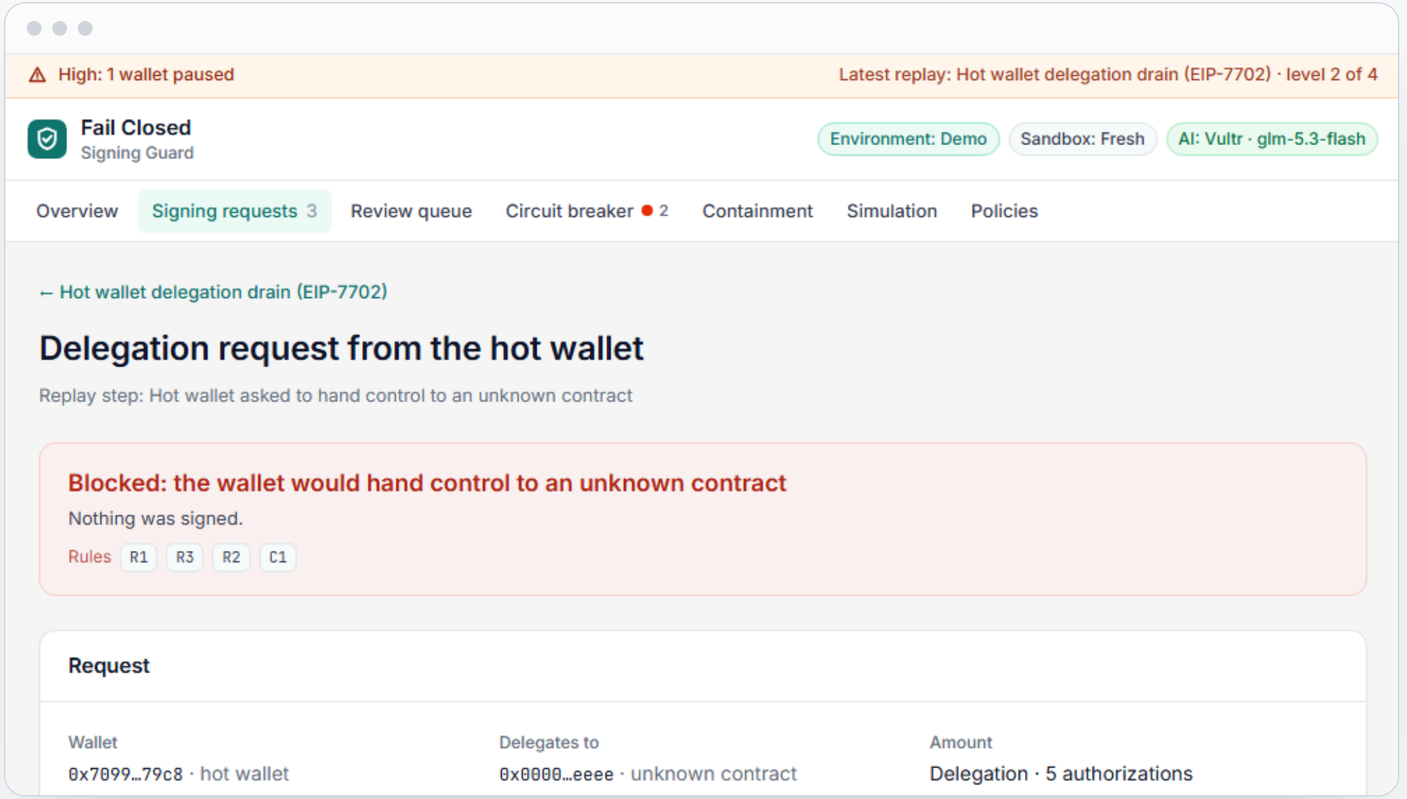
A prototype of that decision logic: fixed rules decide, unusual requests run in an isolated sandbox, every failure fails closed, a scoped circuit breaker, the AI only explains.

Built in this prototype

- Fixed rules decide; the AI explains and may ask for a deeper check
- Isolated sandbox: state diff and a delegation probe
- Fails closed on every error, timeout or empty result
- Scoped circuit breaker: address, wallet, chain, global
- Approval bound to the request hash; the signer checks it

DEMO 1 · HOT WALLET DELEGATION DRAIN

Blocked. Nothing was signed.

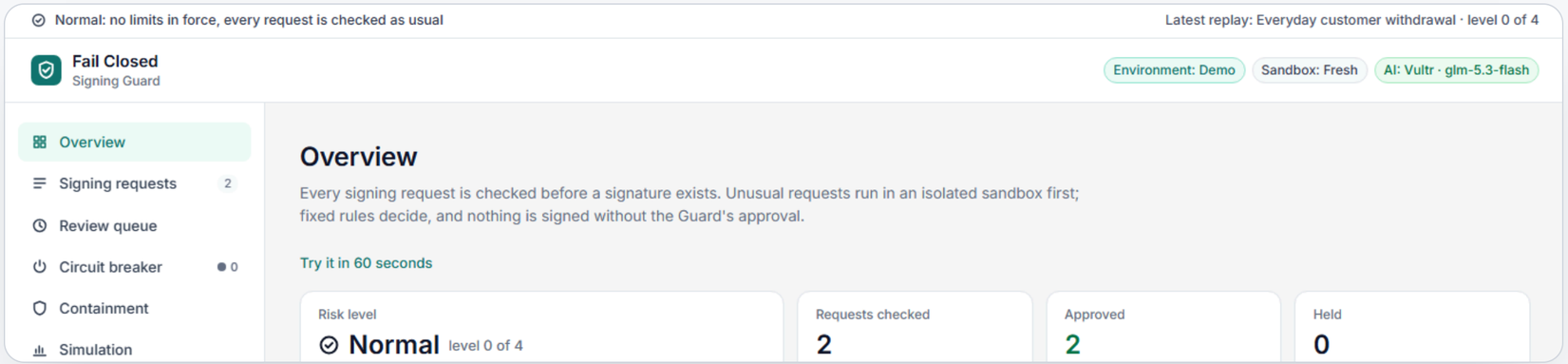


- The sandbox shows the wallet would be **handed to an unknown contract**.
- A test call as the attacker **moves 100 ETH** out (fake funds, in the sandbox).
- No approval, so **the signer refuses**. The wallet is paused.
- The AI (Vultr **glm-5.3**) explains afterwards, in plain words: *"This is a classic pattern for handing over control of a hot wallet, not an ordinary withdrawal."* It did not decide.

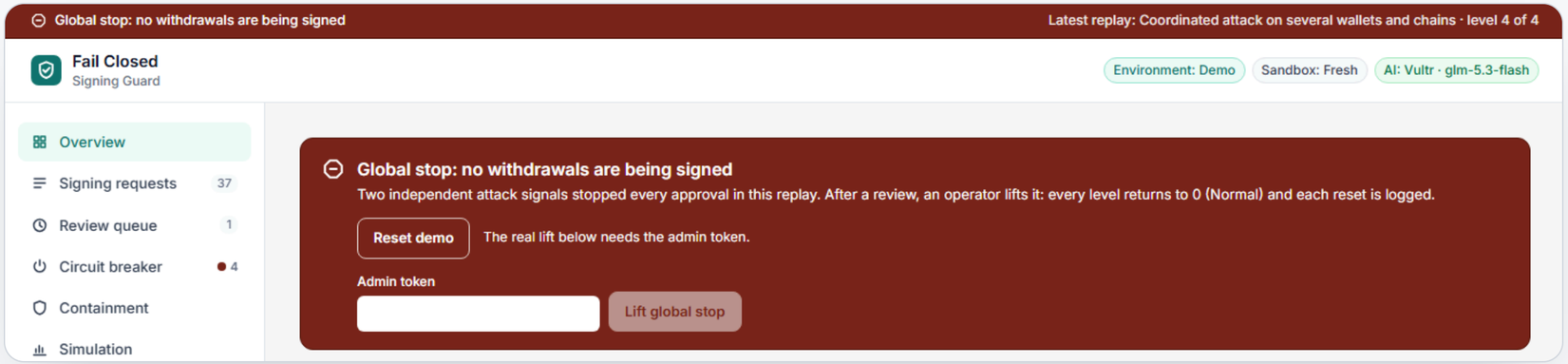
DEMO 2 · COORDINATED ATTACK

The risk level climbs 0 → 4, one narrow step at a time.

Before: level 0, Normal



After the attack: level 4, Global stop



- Level 1 freezes **one address**.
- Level 2 pauses **one wallet**.
- Level 3: **one chain** in safe mode.
- Customers outside the stopped scope keep being paid until level 4.
- Level 4 needs **two independent signals**. Only an operator lifts it, with the admin token.

MEASURED RESULTS

500 normal withdrawals + 1 hidden attack

Attack stopped	**yes, in all 4 setups**
Normal customers refused	**0 of 500**
Fast path (plain payout to a known customer)	**3 ms median (warm setup)**
Slow path (full sandbox check)	**1.3 s median (warm) · 2.4 s (fresh)**
Same check with the real Vultr model (49 + 1)	**attack stopped · 0 refused · 0 rate limits**
Red team: hostile code inside the sandbox	**7 of 7 contained**

Measured on a laptop (Windows 11, 16 CPUs, Docker Desktop), 6 Nov 2026. Rows 1–4: backend/cli.py loadtest, mock AI (800 ms per call), file docs/results/loadtest-20261106-111314.md. Row 5: real Vultr glm-5.3-flash + glm-5.3, file docs/results/live/loadtest-20261106-111340.md. Row 6: backend/cli.py redteam, file docs/results/redteam-20261106.txt. The Vultr VM was blocked by account review, so all numbers are from the laptop.

BUSINESS VALUE

Protect the wallets without stopping normal customers.

- Everyday payouts to known customers are approved **in milliseconds**.
- New destinations and unusual requests get a **full simulation** first, in about 1–3 seconds.
- Large or odd payouts go to **a person**, not to a blanket stop.
- The breaker stops **as little as possible**: address, then wallet, then chain.
- In our load test: **0 of 500** normal customers refused (laptop, mock AI, 6 Nov 2026).

Honest limits

- A hackathon prototype: local test chain, fake funds.
- It does not stop the first break-in. It stops the break-in from getting a signature.
- If an attacker takes over the Guard or its key, the protection is gone: run it separately.
- The Vultr VM was blocked by account review, so the demo and all numbers are from a laptop.

What's next

- Plug into one real signer hook (e.g. an MPC or HSM co-signer callback).
- Check every signature against an approved withdrawal in the exchange ledger, from a separate trust domain.
- Deny unknown operation types on withdrawal wallets (e.g. EIP-7702 delegation).
- Rules for deposit-address sweeps.
- A pre-approved path to revoke a bad EIP-7702 delegation that the breaker never blocks.

No approval, no signature.